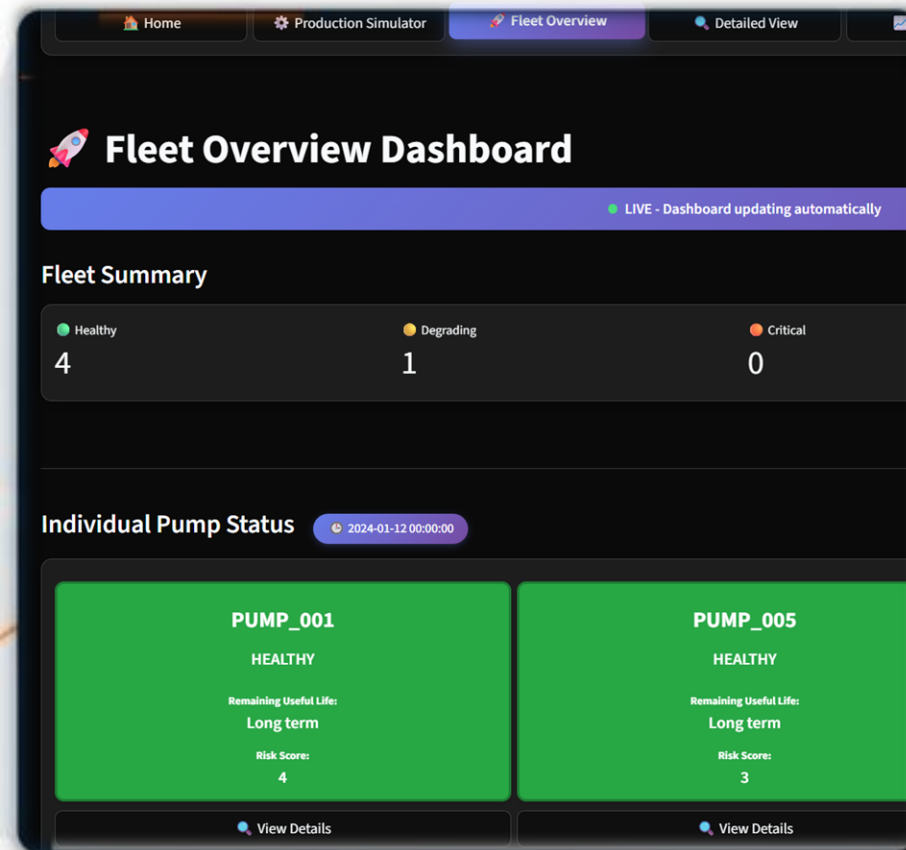
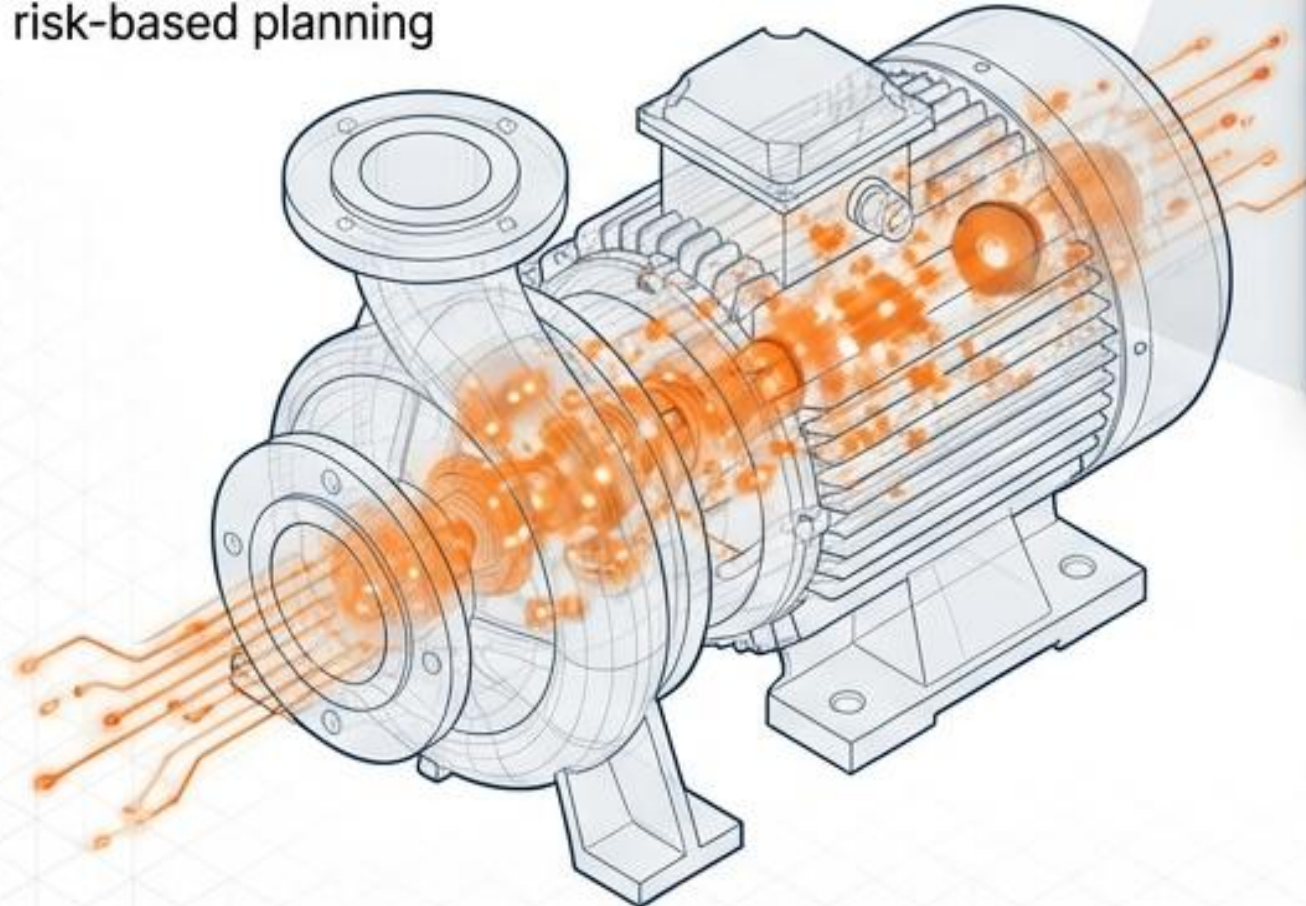


TwinPulse

ML-Based Digital Twin Framework for AI Driven Asset Prognosis | Case Study: Industrial Centrifugal Pumps

THE PROBLEM

- Reactive maintenance leads to cascading production disruptions
- Sudden pump failures cause unrecoverable downtime costs
- Lack of lifecycle visibility prevents risk-based planning



WHAT WE BUILT

- End-to-end framework monitoring 15 industrial pumps
- Processed ~65,000 rows of physics-informed telemetry
- Hybrid ML classifier and deterministic risk engine
- Interactive 'What-If' maintenance simulation sandbox

TARGET USERS

- Plant Operations Heads | Maintenance Managers | Reliability Engineers

OUR SOLUTION

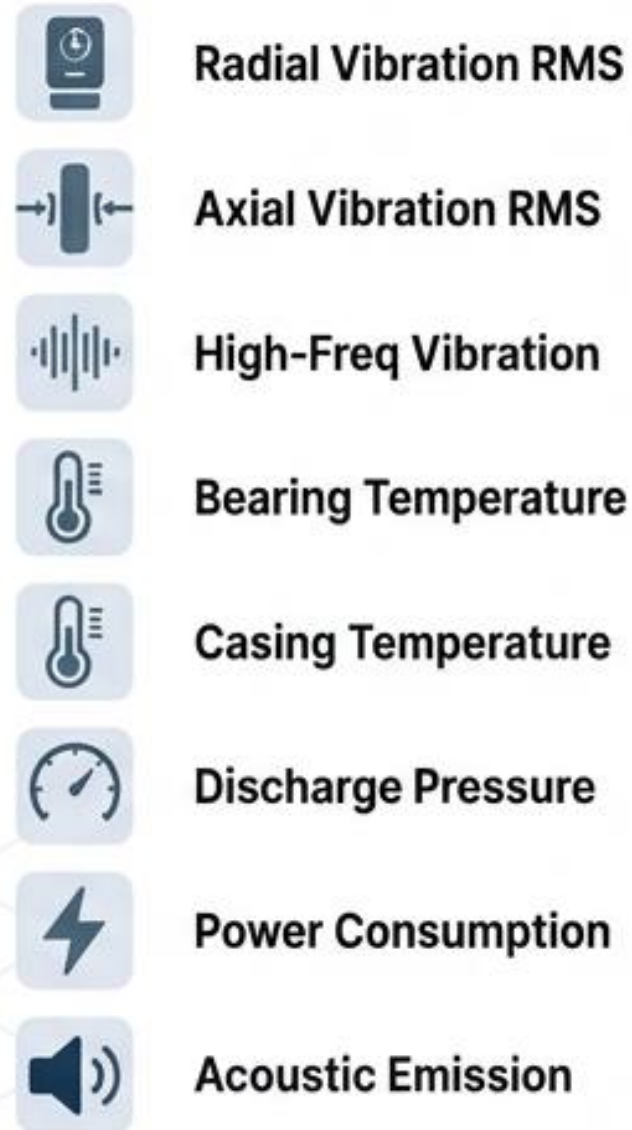
- **Digital Twin Architecture:** Dynamic virtual representation of asset health
- **Predictive Maintenance:** Early failure detection (5–30 day lead time)

TEAM MEMBERS

Shivani Singh	Sunidhii Sharma	Parth Shahi
Aryan Jain	Pratyush Mathur	Abhinav Ranjan

From Sensor Telemetry to Intelligent Decision Support

DATA INGESTION



ANALYTICAL LAYERS

Layer 1: Feature Engineering

Rolling windows (48h/168h), linear slopes, variability metrics

Layer 2: ML Health Classifier (Detection)

Soft-voting ensemble: Random Forest, LightGBM, Logistic Regression
SMOTE balancing and monotonic health latching

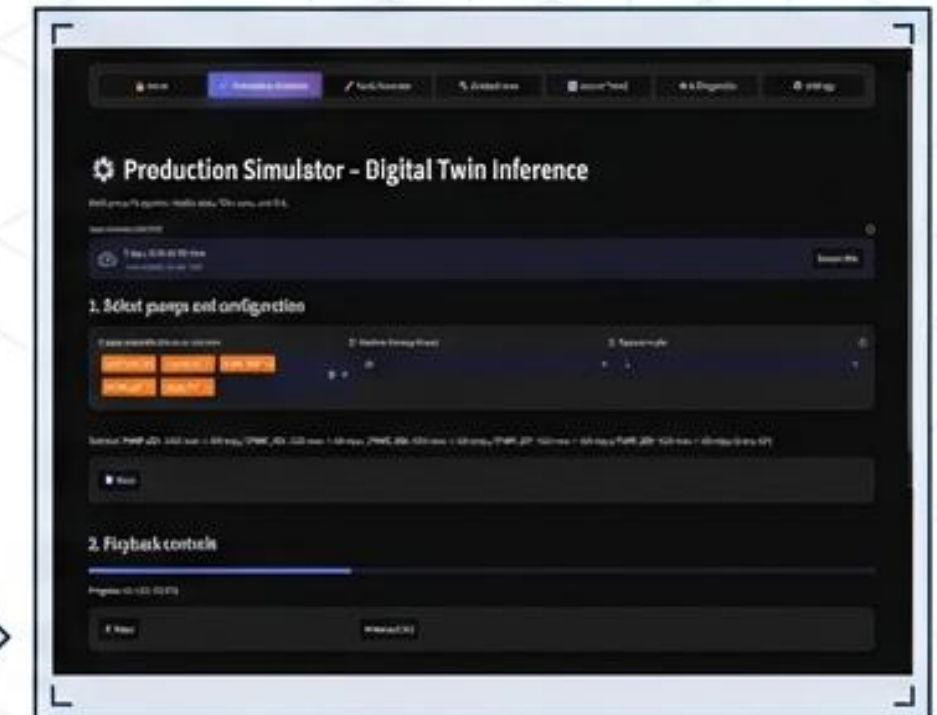
Layer 3: Deterministic Risk Engine (Logic)

Non-ML Layer: Severity-weighted scoring with EWMA smoothing
Captures trend acceleration and dwell-time persistence

Layer 4: RUL Prognostics (Forecast)

Gradient Boosting Regressor with failure clamp (RUL=0)
Enforces monotonic decreasing lifecycle trajectories

OPERATIONAL OUTPUT



Layer 5: Digital Twin Sandbox

Forked baseline vs. intervention simulation

Layer 6: Agentic AI Diagnostic

LLM-driven natural language reasoning and explanation

Technology, Differentiation & Product Value

FULL TECH ECOSYSTEM

Data & Engineering:

Python | Pandas | NumPy

ML & Modeling:

Scikit-learn | LightGBM | Gradient Boosting | SMOTE

Deterministic Modeling:

Custom Risk Engine | EWMA Smoothing Logic

Digital Twin & AI:

Streaming Simulator | State Isolation | LM Studio (Local LLM)

Deployment:

Streamlit Dashboard | Serialized Model Artifacts

STRATEGIC DIFFERENTIATION

Why TwinPulse?

- ✓ **End-to-End Architecture:** From raw sensor data to decision support
- ✓ **Hybrid Logic:** Combines ML probability with deterministic risk rules
- ✓ **Physics-Constrained:** Enforces realistic, monotonic degradation curves
- ✓ **Streaming-Ready:** Simulates real-time inference without data leakage
- ✓ **Interactive Sandbox:** Allows 'What-if' maintenance intervention testing
- ✓ **Agentic AI Layer:** Translates complex data into plain-English diagnostics

BUSINESS VALUE

Operational ROI

5–30 Day Early Warning

Pre-empt catastrophic failures

Downtime Reduction

Eliminate unplanned production stops

Resource Optimization

Transition to condition-based maintenance

Fleet Visibility

Risk-prioritized view of all assets

TwinPulse transforms industrial pump telemetry into predictive operational intelligence.